

Qualidade e Teste de Software - JCRTQSW

Lista de Exercícios 11

- 1) Realizar o teste de mutantes nos trechos de dos códigos apresentados a seguir.
- 2) Apresentar o relatório de mutantes gerado pelo Pitest

mvn test-compile org.pitest:pitest-maven:mutationCoverage

```
void compute(int x, int y) {  
    if (y == 0)  
        System.out.println ("y is zero");  
    else if (x == 0)  
        System.out.println ("x is zero");  
    else {  
        for (int i = 1; i <= x; i++) {  
            if (i % y == 0)  
                System.out.println (i);  
        }  
    }  
}
```

a)

```
boolean lookup(int vector[], int key, int n) {  
    int medium;  
    int low = 0;  
    int high = n;  
    while (low <= high) {  
        medium = (low + high)/2;  
        if (vector[medium] == key) { // value found  
            return medium;  
        }  
        else {  
            if (vector[medium] < key)  
                low := medium + 1;  
            else
```

```
        high := medium - 1;

    }

}

return -1;

}
```

b)

```
public enum ErrorCodes {
    EACCES(0), ENODEV(1), ENOENT(2);
    int _errorn;
    ErrorCodes(int e) { _errorn = e; }
    public int getErrorNum() { return _errorn; }
}

public class CoverageCodeExample {
    public void example1(boolean condition, String string1, ErrorCodes errno, char input[])
    {
        int p[] = null;
        char string2[] = null;
        String message[] = new String[3];
        int i = 0;
        if (condition) {
            p = new int[1];
            p[0] = 1;
        }
        p[1] = 0;

        if (condition || string1.equals(string2))
            System.out.println("They are equal");

        message[ErrorCodes.EACCES.getErrorNum()] = "Permission denied";
        message[ErrorCodes.ENODEV.getErrorNum()] = "No such device";
        message[ErrorCodes.ENOENT.getErrorNum()] = "No such file or directory";

        switch (errno.getErrorNum()) {
```

```
        case 0:
        case 1:
        case 2:
            System.out.printf("%s\n", message[errno.getErrorNum()]);
            break;
    }

    char output[] = new char[100];
    for (i = 0; i <= output.length; i++) {
        output[i] = input[i];
        if (input[i] == '\\0')
            break;
    }
    i = 0;

    do {
        output[i] = input[i];
    } while (input[i] != '\\0');
}
}
```

c)

```
public Integer m(int i, Integer obj) {
    int s = 0;

    if (obj == null)
        return null;

    if (i % 2 == 1)
        i++;

    for(; i > 0; i--)
        s += obj.intValue();

    return s;
}
```

d)

```
void doSomething(int x, double vec[]) {  
    int y = 0;  
    if (x > 0) y = 1;  
    if (x > 100 && y != 0)  
        System.out.println("less than 100");  
    else  
        y = vec[0];  
    switch (y) {  
        case 1:  
            f(vec);  
            break;  
        case 2:  
            g(vec);  
            break;  
    }  
    ff(vec);  
}
```

e)

```
void contains(int x, int a, int b) {  
    if(a <= b) {  
        if(a <= x && x <= b)  
            System.outprint.ln("x is an interior point");  
        if(x <= a || x >= b)  
            System.outprint.ln("x is outside the interval");  
    } else  
        System.outprint.ln("the interval is empty");  
}
```